A PROPERTY LOCAL ON THE SOURCE IS PRESERVED BY FLAT, SYNTOMIC, ETALE AND OPEN PRECOMPOSITION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Localness of a property of morphisms on the source for a topology. Same chapter, the definition of tau localness on the source, named in the statement at lines 6876-6877. Supplies the axiom that transfers the property along a covering family of the source.
- A single flat (or fppf, syntomic, etale, open immersion) morphism forms a covering family. Standard, used at proof lines 6885-6886. Supplies the covering family into which a is fitted.

Lemma 0.1. *Let $\tau \in \{fpqc, fppf, syntomic, smooth, étale, Zariski\}$. Let $\mathcal{P}$ be a property of morphisms which is τ local on the source. Let $f : X \to Y$ have property $\mathcal{P}$. For any morphism $a : X' \to X$ which is flat, resp. flat and locally of finite presentation, resp. syntomic, resp. étale, resp. an open immersion, the composition $f \circ a : X' \to Y$ has property $\mathcal{P}$.*

Proof. This is true because we can fit $X' \to X$ into a family of morphisms which forms a τ-covering. □

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